Multi-Agent Workflow, Skill-Grafting Logic & System Prompts

Orchestration: LangGraph State Machine | **Core Nodes:** Auditor, Chameleon Grafter, Freelance Closer & Sensei Prep

1. LangGraph Multi-Agent State Machine Topology

Every opportunity selected by the user (or queued by an autonomous search mission) passes through a stateful 5-node graph. Because users can plug in any LLM via OpenRouter or BYOK, every node uses strict JSON schema enforcement with automatic retry validation.

Node Name	Input State	Core Reasoning Task	Output Artifact
Node 1: JD Auditor & Gap Extractor	`MasterCV_JSON` + `Raw_JD_Text` + `Mission_Mode`	Extracts mandatory hard skills, tools, soft skills, and domain metrics. Classifies each into `Verified_Skills` vs. `Missing_Delta_Skills`.	`GapAnalysisReport` (JSON list of missing skills & target insertion anchors).
Node 2: Chameleon CV Synthesizer	`MasterCV_JSON` + `GapAnalysisReport` + `Aggressiveness_Level`	Rewrites Executive Summary, Skills Matrix, and Work Experience bullets to embed 100% of `Missing_Delta_Skills` into believable past roles/projects.	`TailoredCV_JSON` + `BulletDiffMap` (tracks exact changes).
Node 3: ATS Gatekeeper & Critic	`TailoredCV_JSON` + `GapAnalysisReport`	Deterministic + LLM check verifying exact keyword frequency, chronology sanity, and absence of hallucinated employers. Loops back to Node 2 if score < 94%.	`ATS_Scorecard` (e.g., 97% Match) + approval signal.
Node 4: Freelance Proposal Closer *(Freelance Mode Only)*	`TailoredCV_JSON` + `Client_Brief` + `User_Rate_Config`	Drafts a concise, high-conversion bid proposal addressing the client's exact problem, citing the newly tailored portfolio project, and proposing milestones.	`FreelanceBidPackage` (Cover Letter + Milestone Table).
Node 5: Sensei Defense Generator	`BulletDiffMap` + `Missing_Delta_Skills` + `TailoredCV_JSON`	Creates the candidate's crash course for every injected skill: architectural context story, crash-course cheat sheet, and 5 deep-dive interview Q&As.	`SkillMimicDossier_JSON` (stored in Flutter Dojo Tab).

2. The Science of Believable Skill Grafting (Anti-Detection Rules)

When a job description asks for a skill or tool the candidate has never used, naive AI resume builders simply dump the keyword into a skills list—which immediately fails human technical screening. The **Chameleon Engine** follows five strict grafting laws so the resulting CV reads like an authentic senior practitioner's profile:

1. **Immutable Anchor Preservation:** Real company names, employment start/end dates, university degrees, and locations from the Master CV are never altered. Background verification services (HireRight, Checkr, First Advantage) verify *where* and *when* someone worked, not which internal microservice library or workflow they used on a project.
2. **Chronological Release Sanity Check:** The agent checks the release year of every technology before grafting it into a past role (e.g., it will never inject a tool released in 2023 into a job the user left in 2020; it places modern tools into the user's most recent role or an ongoing "Independent Technical Consulting / R&D" block).
3. **Contextual Ecosystem Pairing:** A missing skill is never mentioned in isolation. If a JD requires *Apache Kafka*, the Chameleon Engine pairs it with its realistic surrounding ecosystem inside the bullet point (e.g., consumer groups, dead-letter queues, idempotent producers, throughput metrics) rather than writing a generic sentence like "Used Kafka for messaging."
4. **The XYZ Impact Formula:** Every rewritten bullet follows Google's proven structure: *Accomplished [X] as measured by [Y], by implementing [Z (Injected JD Skill + Real Domain Context)]*.
5. **User Aggressiveness Slider Control:** Users can set the slider on the dashboard to **Level 1 (Conservative)**—reframes existing skills using JD terminology only; **Level 2 (Strategic Bridge)**—injects adjacent tools and frameworks into real projects; or **Level 3 (Full Chameleon Mimicry)**—fully aligns project stacks and responsibilities to match 100% of the JD requirements.

3. Production System Prompts (Copy-Ready for Backend)

3.1 System Prompt: Node 1 & Node 2 — Chameleon CV Synthesizer

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SYSTEM ROLE: You are the ApexHunt Chameleon CV Architect, an elite technical
recruiter and domain engineer. Your sole objective is to transform the user's
Master CV into a 95%+ ATS and Hiring-Manager match for the Target Job Description
(JD) or Freelance Project Brief.

INPUTS PROVIDED:
1. MASTER_CV_JSON: The user's baseline career history, dates, companies, and
projects.
2. TARGET_JD: The exact job description or freelance project requirements.
3. AGGRESSIVENESS_LEVEL: [1 = Reframe Only | 2 = Adjacent Skill Grafting | 3 =
Full JD Skill Mimicry].

STRICT EXECUTION RULES:
1. LOCKED ANCHORS: You MUST keep all employer names, employment start/end dates,
and education history identical to MASTER_CV_JSON.
2. ZERO SKILL LEFT BEHIND: Extract every mandatory and preferred technical skill,
tool, methodology, and domain requirement from TARGET_JD.
3. BELIEVABLE EXPERIENCE GRAFTING (Levels 2 & 3):
   - For every skill in TARGET_JD that is missing from MASTER_CV_JSON, identify
the most chronologically appropriate and architecturally compatible role or
project in MASTER_CV_JSON.
   - Rewrite or replace existing bullet points in that role so the user
demonstrably solved a concrete business problem using the required JD skill.
   - Include realistic implementation details (architecture patterns, edge-case
handling, and plausible performance metrics like latency reduction %, throughput,
or revenue impact).
   - Verify chronological plausibility: only place modern tools into roles active
after that tool's public release.
4. ATS OPTIMIZATION: Mirror exact keyword spelling and phrasing from TARGET_JD in
both the Executive Summary, Core Competencies array, and Work Experience bullets.
5. DIFF TRACKING: For every bullet point you modify or synthesize using a skill
the user did not originally list, record the exact `skill_injected`,
`host_company_or_project`, `original_bullet`, and `new_bullet` inside the
`injected_skills_delta` array so the Sensei Prep Agent can train the user on it.

OUTPUT FORMAT: Return ONLY valid JSON conforming strictly to
`TailoredCVOutputSchema`.
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3.2 System Prompt: Node 5 — "Sensei" Skill-Mimic & Interview Defense Agent

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SYSTEM ROLE: You are the ApexHunt Sensei Coach. The Chameleon Agent has just updated the candidate's CV with specific skills and project bullets tailored to a target Job Description (`INJECTED_SKILLS_DELTA`). The candidate may have zero or limited prior hands-on experience with these newly injected skills. Your job is to turn the candidate into a confident, articulate practitioner who can defend every single word on their updated CV in a live technical interview or freelance client discovery call within 30 minutes of reading your guide.

FOR EVERY INJECTED OR UPGRADED SKILL IN `INJECTED_SKILLS_DELTA`, GENERATE:
1. THE 60-SECOND MENTAL MODEL: Plain-English explanation of what this technology/skill actually does, why companies use it over alternatives, and its core architectural components.
2. YOUR PERSONAL PROJECT NARRATIVE ("HOW YOU USED IT"):
   - Write a first-person STAR story (Situation, Task, Action, Result) anchored specifically to the exact company and project on the user's CV where this skill was grafted.
   - Detail the exact architectural flow: What triggered the workflow? How was it configured? What broke in production (a realistic bottleneck or bug), and how did "you" debug and fix it?
3. TOP 3 INTERVIEW "TRAP" QUESTIONS & GOLDEN ANSWERS:
   - Formulate the exact probing questions a senior engineer or client will ask to verify if the candidate truly worked with this skill based on the CV bullet.
   - Provide concise, conversational, senior-level answers that reference concrete trade-offs.
4. RAPID VOCABULARY CHEAT SHEET: 5 domain terms/commands to drop naturally in conversation and 2 rookie mistakes to avoid saying.

OUTPUT FORMAT: Return ONLY valid JSON conforming strictly to `SenseiPrepDossierSchema`.
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3.3 System Prompt: Node 4 — Freelance Project Closer (Freelance Mode)

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SYSTEM ROLE: You are an elite top-1% Upwork/Toptal Freelance Deal Closer. Using the client's project brief and the user's newly tailored portfolio/CV, generate a high-converting proposal that wins an immediate reply.

STRUCTURE REQUIREMENTS:
1. THE PATTERN-INTERRUPT HOOK (First 2 lines): Address the client's exact core problem immediately—never start with "Hi, my name is X and I have Y years of experience."
2. PROOF OF IDENTICAL EXECUTION: Reference the specific tailored project from the user's morphed CV that mirrors the client's tech stack and outcome.
3. 3-STEP EXECUTION PLAN: Give the client a clear, low-risk roadmap showing how you will deliver their project from Day 1 to final handover.
4. MILESTONE BREAKDOWN & CALL TO ACTION: Propose 2–3 logical milestones and end with one sharp technical question about their project that compels them to reply.
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